

SchoolDiary

Parent–Teacher Communication App for a School Group

Case ID	111
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Academic Programme	B.Tech Computer Science & Engineering (2025–2029)
Curriculum Subject	Software Engineering & Project Management — Semester III
Pilot Scope	2 Schools · 2,300 Parents · ~107 Teachers (16.4% Group Share)
Engineering Budget	3 Developers · 68 Person-Days · 27 Working Days Scheduled
Deliverables Included	Full IEEE 830 SRS, UML Suite, CPM Plan, Estimation, Scope, IEEE 829 Tests, Risk Register, Closure Note

Executive Problem Statement

A group of 12 schools communicates with parents through paper diaries, circulars and dozens of WhatsApp groups run by individual teachers. Parents miss critical circulars, teachers receive messages late at night, and management maintains zero centralized audit trail. The group commissioned SchoolDiary to provide reliable circulars, homework distribution, 15-minute attendance alerts, fee reminders, and structured two-way messaging strictly constrained within school hours.

14,000 Parents	650 Teachers	12 Schools	2,300 Pilot Parents	16.4% Pilot Share
68 Person-Days	27 Working Days	Team of 3	90% Monthly Active Target	0 Late Messages Target

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SECTION 01

Software Requirements Specification (SRS)

IEEE 830-Style Specification · Baseline Version 1.1

1.1 Purpose, Scope & Baseline Group Figures

This SRS formally specifies the requirements for SchoolDiary across 12 schools, establishing the contractual baseline for the UML architecture, critical-path project schedule, test plan, and pilot rollout. SchoolDiary is architected as an Android application for parents and teachers coupled with a secure web administration console.

Metric / Constraint	Baseline Value	Exact Mathematical Derivation & Operational Context
Total Parent Base	14,000 parents	Total registered parent body across all 12 institutions (given).
Total Faculty Staff	650 teachers	Total instructional staff across the group (given).
Institutions in Scope	12 schools	Group footprint; pilot restricted to 2 institutions (given).
Pilot Parent Cohort	2,300 parents	Active parent group across the 2 pilot schools (given).
Pilot Share of Parents	16.43% (≈ 16.4%)	$2,300 \div 14,000 = 0.1642857\dots$ exactly 16.43% of total population.
Pilot Faculty Allocation	~107 teachers	$650 \times (2,300 \div 14,000) = 106.78 \approx 107$ teachers (proportional).
Monthly Active Target (Pilot)	2,070 parents	$90\% \times 2,300$ pilot parents = 2,070 parents active monthly.
Full Rollout Active Target	12,600 parents	$90\% \times 14,000$ group parents = 12,600 parents active monthly.
Late Messaging SLA Target	0 messages	Teacher messages delivered after 20:00 reduced to absolute zero.

1.2 Users & System Constraints

User Class	Primary Operational Need	Strict Access Boundary & Data Scope
Parent	Receive notices, homework, absence alerts; message teachers	Strictly isolated to their own enrolled child's records (NFR-03)
Teacher	Publish circulars/homework; mark attendance; message without night spam	Restricted strictly to mapped classrooms and assigned subjects
School Admin	Approve school notices; manage rosters and staff mappings	Scoped strictly to their assigned institution
Group Management	Audit group-wide communications and review compliance rates	Read-only access across all institutions; no direct messaging capability
Principal	Adjudicate out-of-hours urgent teacher messaging approval requests	Institutional authority for emergency exception overrides

Architectural Constraints: Android 8.0+ clients with ≥2 GB RAM; web administration console for desktop browsers; Indian Standard Time (IST, UTC+05:30) at minute-level granularity; English UI for pilot; strict exclusion of payment processing; 3-person engineering team with 68 person-day bottom-up budget; end-to-end TLS 1.2+ and AES-256 encryption at rest.

1.3 Functional Requirements (FR-01 to FR-10)

ID	Requirement Statement	Priority	Measurable Acceptance Check & Test Oracle
FR-01	Role-scoped authentication for parent, teacher, admin, management, principal	Must	Authenticate credentials per role; deny cross-class and cross-child unauthorized access requests (TC-01)

ID	Requirement Statement	Priority	Measurable Acceptance Check & Test Oracle
FR-02	Circular publishing: class-level immediate; school-level requires admin approval	Must	Class notice posts immediately; school-level notice remains in PENDING state until admin approves (TC-02)
FR-03	Per-parent read receipts: delivered and read timestamps with aggregate read %	Must	Simulate 10 parents: 6 open notice -> system displays 60% read and lists 4 unread parents by name (TC-03)
FR-04	Homework module: post tasks with due dates; track parent acknowledgements	Should	Post class homework; parent acknowledgement persists and reports to faculty dashboard (TC-04)
FR-05	Attendance alerts: notify parents within 15 min of roll-call submission	Must	Mark student absent at 09:00 roll call -> push notification delivered to parent device by 09:15 (TC-05)
FR-06	Fee reminders: automated triggers at D-7 and D-1 before due dates (No Payment)	Should	Reminders fire at D-7 and D-1 only; interface contains zero payment gateway or link (TC-06)
FR-07	Two-way parent-teacher threads per child with mutual phone number masking	Must	Parent messages class teacher; thread isolates by child; phone numbers masked as XXXX-XXXX (TC-07)
FR-08	Messaging-hours enforcement engine (07:00–19:59 window and queueing)	Must	Execute all boundary and decision table tests; block or queue out-of-window attempts (TC-08)
FR-09	Admin portal: bulk CSV roster import with row-level validation; approval queue	Must	Import 100 rows with 1 error: 99 commit, 1 reports exact line number and failure reason (TC-09)
FR-10	Communication audit log and compliance reports (Active Parents & Late Messages)	Should	Audit log searchable by date/sender; reports output exact numerator, denominator, and % (TC-10)

1.4 Measurable Non-Functional Requirements (NFR-01 to NFR-06)

ID	Quality Attribute	Specification (Zero Ambiguity)	Verification Method
NFR-01	Delivery Timeliness	≥99% of circulars, homework and alerts reach push gateway within 5 min; attendance alerts reach device within 15 min of roll-call submission	10,000 synthetic message load test + gateway timestamp audit
NFR-02	Delivery Integrity	0 lost or duplicated messages across 10,000 retried deliveries via idempotency keys; read receipts stored within 60 s of user view	Chaos-injection retry test harness and database consistency check
NFR-03	Access Control Privacy	100% pass rate on role-boundary penetration tests; zero cross-tenant, cross-class, or cross-child leakage	Automated security test suite executing 500 boundary test permutations
NFR-04	Data Protection	100% TLS 1.2+ in transit; AES-256 at rest; phone numbers masked in all user UI; student data scrubbed 12 months after exit	Cryptographic configuration review, database dumps, and code audit
NFR-05	System Usability	≥90% of 20 test parents read notice and reply in ≤3 taps from home; operates smoothly on Android 8+ (2 GB RAM); Class-5 reading level	Structured usability laboratory evaluation with 20 representative parents
NFR-06	High Availability	≥99.5% uptime during operational core hours (06:30–21:00 IST Monday–Saturday)	Automated synthetic probe checks every 60 seconds from external vantage points

1.5 Messaging-Hours & Approval Rules (FR-08)

Governing Messaging Rules (Indian Standard Time, Minute-Level Oracle):

- Faculty Sending Window:** 07:00–19:59 IST inclusive. Teacher messages dispatched immediately.

2. **Routine Out-of-Hours Messaging:** Between 20:00 and 06:59 IST, routine messages are blocked and queued for automated dispatch at 07:00 next school day.
3. **Urgent Messaging Exception:** Out-of-hours messages marked Urgent trigger an instant alert to the Principal. If approved, sent immediately; if rejected or undecided by 07:00, released at 07:00.
4. **Parent Submission Flexibility:** Parents may send messages 24/7. Out-of-hours parent messages are held for faculty review at 07:00, and parent receives an instant automated auto-reply.
5. **Zero Late Messaging Target:** Teacher-initiated routine messages delivered after 20:00 must equal zero.

1.6 MoSCoW Priorities & Traceability Matrix

MoSCoW Tier	Requirements Allocated	Strategic Rationale & Trade-Off Analysis
Must Have	FR-01, FR-02, FR-03, FR-05, FR-07, FR-08, FR-09	Non-negotiable core: solves the primary brief problems (missed notices, late messages, no records).
Should Have	FR-04, FR-06, FR-10	High-value items: homework posting, fee reminders, and compliance audit reporting. Pilot viable temporarily without them.
Could Have (Backlog)	C-01 Multilingual UI (Hindi/Marathi), C-02 SMS Fallback, C-03 PTM Slot Booking	Valuable extensions. C-02 SMS fallback is prioritised as contingency if parent app adoption lags below target.
Won't Have (Release 1)	Payment Gateway, Video Calls, Student Accounts, WhatsApp Integration	Excluded to eliminate security liabilities, telecom fees, minor data risks, and channel circumvention.

Requirements Traceability Matrix (RTM)

Requirement	System Test Case	Unit / Integration / Boundary Test Verification
FR-01 Accounts & Scoping	TC-01	TC-EC-05, TC-EC-06; Automated role-boundary test suite (NFR-03)
FR-02 Circulars & Approval	TC-02	Decision table rules TC-DT-R8 to R13; TC-EC-10, TC-EC-11
FR-03 Read Receipts	TC-03	Receipt latency benchmark harness (NFR-02); database uniqueness test
FR-04 Homework & Acknowledge	TC-04	Class filtering test; parent acknowledgement persistence check
FR-05 Attendance Alerts	TC-05	End-to-end roll call timer benchmark; 15-minute SLA check (NFR-01)
FR-06 Fee Reminders (No Pay)	TC-06	D-7 and D-1 scheduler tests; static security scan for payment endpoints
FR-07 Two-Way Messaging	TC-07	Thread isolation tests; TC-BVA-10 to 18; phone masking verification
FR-08 Messaging Hours Engine	TC-08	TC-BVA-01 to 18; TC-DT-R1 to R13; TC-EC-01 to 09
FR-09 Admin Roster Management	TC-09	100-row CSV parser test; line-error reporting validation
FR-10 Audit Log & Compliance	TC-10	Audit log query benchmarks; monthly active & after-8 p.m. formula verification

SECTION 02

UML Software Design Package

2.1 Use-Case Model & System Actors

The use-case model delineates the exact operational boundary of SchoolDiary. Actors sit outside the system boundary: Parent, Teacher, School Admin, Principal, Group Management, and the external Push Gateway (secondary system actor).

Actor	Primary Use Cases Executed	Relationship Details («include» / «extend»)
Parent	View & acknowledge notice; view homework; send message to teacher	Send message «include» Sign in; Auto-reply «extend» Send message [parent after hours]
Teacher	Publish notice; track read receipts; post homework; mark attendance; message parent	Send message «include» Check hours; Send urgent «extend» Send message; Notice «include» Push
School Admin	Approve/reject school notices; manage rosters and mappings; view queue	Publish school notice «include» Admin approval; Bulk CSV import «include» Sign in
Principal	Adjudicate emergency out-of-hours teacher messages	Approve urgent message «extend» Send urgent message [outside window]
Management	View central audit log; inspect monthly active and late-message reports	View reports «include» Sign in; Read-only scoping strictly enforced
Push Gateway	Deliver push notifications to mobile client devices (secondary actor)	Invoked asynchronously by NotificationDispatcher; accepts idempotency key

2.2 Class Model & Domain Architecture

Domain Class / Enum	Key Structural Attributes & Public Interfaces	Architectural Role & Invariants
School	schoolId, name, principalId, addClass(section)	Institutional root aggregate; encapsulates ClassSections
ClassSection	sectionId, grade, division, classTeacherId, students()	Classroom container; maps faculty to enrolled pupil rosters
Parent	parentId, name, maskedContact, deviceToken, childrenIds	Guardian account; contact masked in all teacher-facing displays
Teacher	teacherId, name, mappedSectionIds, isClassTeacher()	Faculty account; restricted strictly to assigned section boundaries
Notice	noticeId, scope (CLASS/SCHOOL), status, submit(), publish()	Status: DRAFT, PENDING_APPROVAL, QUEUED, PUBLISHED, REJECTED
ReadReceipt	noticeId, parentId, deliveredAt, readAt, markRead()	Composite primary key (noticeId, parentId) prevents duplicate counts
Message	messageId, senderRole, body, urgent, status, scheduledFor	Status: DRAFT, SUBMITTED, QUEUED, PENDING_APPROVAL, SENT, DELIVERED, READ, FAILED
MessagingPolicy	windowStart=07:00, windowEnd=19:59, canSendNow()	Stateless pure rules engine; evaluates (role, time, urgency) -> Decision
NoticeService	compose(), submit(), onApproved(), remindUnread()	Circular orchestration service; publishes NoticePublished domain event

Domain Class / Enum	Key Structural Attributes & Public Interfaces	Architectural Role & Invariants
MessagingService	send(), onParentMessage(), releaseQueued(), retry()	Thread orchestration service; publishes MessageReady domain event
NotificationDispatcher	onNoticePublished(), onMessageReady(), fanOut(), retry()	Shared event consumer; manages retry queue, idempotency, and gateway
PushGateway	push(deviceToken, payload, idempotencyKey): Result	Abstract gateway interface; concrete FcmPushAdapter implements it

2.3 Sequence Diagram: Send Notice & Track Receipts

Illustrates the lifecycle of publishing a circular and capturing receipts. Two alt frames handle approval and window checking.

Step	Origin -> Destination	Invocation / State Transition Details
1	Teacher -> TeacherApp	Compose notice (title, body, attachment, scope=SCHOOL)
2	TeacherApp -> NoticeService	submit(draftNotice)
3	NoticeService	Validate payload size (<=10MB), recipient scopes, author credentials
4	[alt: scope == SCHOOL]	Raise ApprovalRequest -> route to SchoolAdmin portal queue
4a	[opt: Admin Rejects]	SchoolAdmin -> NoticeService: reject(reason) -> Status REJECTED; process halts
4b	[opt: Admin Approves]	SchoolAdmin -> NoticeService: approve(requestId) -> proceed to window check
5	[alt: scope == CLASS]	Bypass admin approval; proceed directly to window check
6	NoticeService -> Policy	canSendNow(role=TEACHER, timestamp=now, urgent=false)
7	[alt: In Window]	Policy -> ALLOW -> Status PUBLISHED; emit NoticePublished(noticeId, audience)
8	[alt: Out of Window]	Policy -> QUEUE_FOR_0700 -> Status QUEUED; emit NoticePublished at 07:00
9	Dispatcher -> Gateway	Fan-out to parent device tokens; push(token, payload, idempotencyKey)
10	[opt: Network Failure]	Gateway returns timeout -> Dispatcher retries (exponential backoff, <=3 attempts)
11	Parent -> ParentApp	Parent opens push notification -> views notice attachment
12	ParentApp -> ReceiptService	record(noticeId, parentId, state=READ) -> persistent timestamp <=60s
13	Teacher -> TeacherApp	Refresh notice analytics -> ReceiptService returns read % and unread names
14	[opt: Unread > 0]	Teacher taps "Remind Unread" -> triggers targeted notification to unread subset

2.4 Activity Diagram: Teacher Sends a Message

Decision Node	Branch Evaluated	Execution Action & System State Transition
Teacher taps Send	—	Initiate transaction; capture IST timestamp and sender credentials
Current time in window?	Yes (07:00–19:59 IST)	Dispatch immediately -> Dispatcher enqueues push -> Write AuditEntry -> End
Current time in window?	No (20:00–06:59 IST)	Evaluate urgency flag: is message marked Urgent?
Urgency flag status	No (Routine Message)	Transition to QUEUED state; display "Queued for 07:00" to teacher; dispatch at 07:00
Urgency flag status	Yes (Urgent Message)	Create ApprovalRequest -> alert Principal with push notification
Principal adjudication	Approved by Principal	Transition to SENT state immediately -> Dispatcher pushes -> log urgent override

Decision Node	Branch Evaluated	Execution Action & System State Transition
Principal adjudication	Rejected OR No decision by 07:00	Transition to QUEUED state -> automatically dispatched at 07:00 next school day

2.5 State Diagram: Message Lifecycle

Initial State	Event / Trigger	Guard Condition	Target State	Lifecycle Meaning
DRAFT	Sender taps Send	—	SUBMITTED	Message validated by client
SUBMITTED	Evaluate sending window	Time in 07:00–19:59 IST	SENT	Immediate gateway dispatch
SUBMITTED	Evaluate sending window	Time outside window & Routine	QUEUED	Held for 07:00 morning release
SUBMITTED	Evaluate sending window	Time outside window & Urgent	PENDING_APPROVAL	Awaiting Principal adjudication
PENDING_APPROVAL	Principal approves	Before 07:00 IST	SENT	Emergency override dispatch
PENDING_APPROVAL	Principal rejects / timeout	Time reaches 07:00 IST	QUEUED	Released with regular morning queue
SENT	Gateway pushes to client	Device confirms receipt	DELIVERED	Push delivered on device screen
DELIVERED	Parent opens message	User interaction recorded	READ	Final terminal state for thread
SENT	Network timeout / error	Retry attempts < 3	FAILED	Enqueued for exponential retry
[START_PARENT]	Parent posts message	Time outside 07:00–19:59	RECEIVED_AFTER_HOURS	Parent initiates after hours
RECEIVED_AFTER_HOURS	Auto-responder executes	Immediate	AUTO_REPLIED	Automated guidance sent to parent
AUTO_REPLIED	Hold message for faculty	Until 07:00 next school day	HELD_FOR_TEACHER	Hidden from teacher notification tray
HELD_FOR_TEACHER	Clock reaches 07:00	Morning release event	DELIVERED_TO_TEACHER	Faculty alerted at start of day

2.6 Cohesion & Loose Coupling Architecture

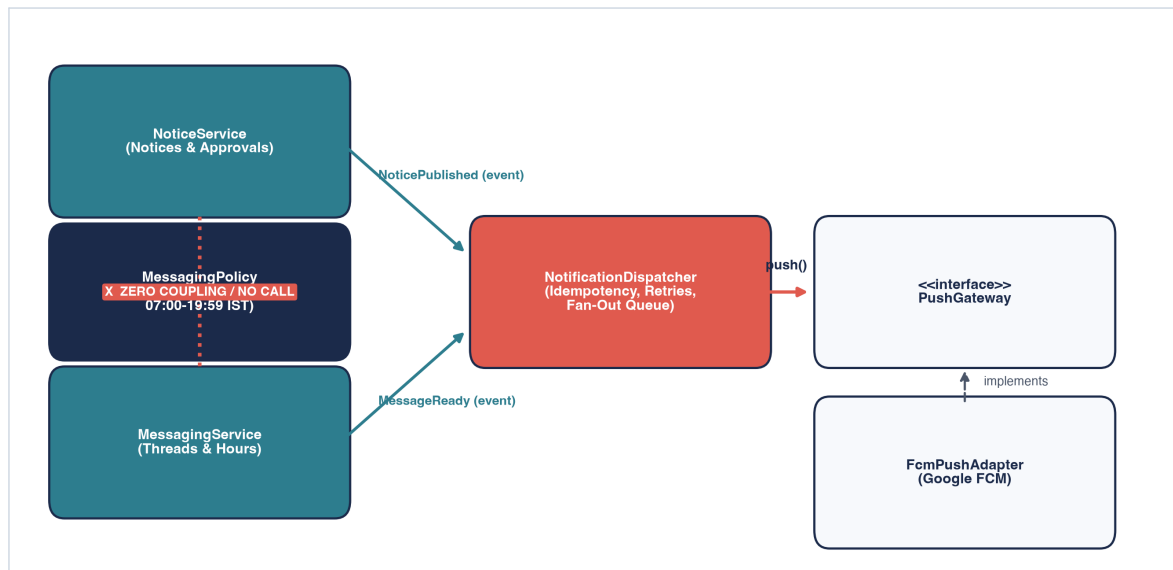


Figure 2.1: Domain Architecture — Zero Direct Coupling Between NoticeService and MessagingService

Architectural Cohesion & Loose Coupling Justification:

- 1. Zero Direct Service Coupling:** NoticeService and MessagingService maintain zero method calls, zero interfaces, and zero foreign keys between them. Circulares evolve with administrative workflows; messaging evolves with parental communication policies.
- 2. Asynchronous Event-Driven Integration:** NoticeService publishes NoticePublished and MessagingService publishes MessageReady. The shared NotificationDispatcher consumes both events asynchronously, centralizing token resolution, idempotency keys, and gateway retry policies.
- 3. Shared Stateless Rules Engine:** Both modules delegate sending window evaluations to MessagingPolicy. This ensures a single, immutable source of truth for the 07:00–19:59 IST schedule without code duplication.

SECTION 02 (CONT.)

2.7 UML Diagram Gallery

This gallery provides the graphical UML views that complement the tabular specifications in Sections 2.1–2.5. Each figure is drawn directly from the corresponding table, so every element can be traced to the SRS (Section 1) and to the test suite (Section 6).

2.7.1 Use-Case Diagram — System Boundary & Actors

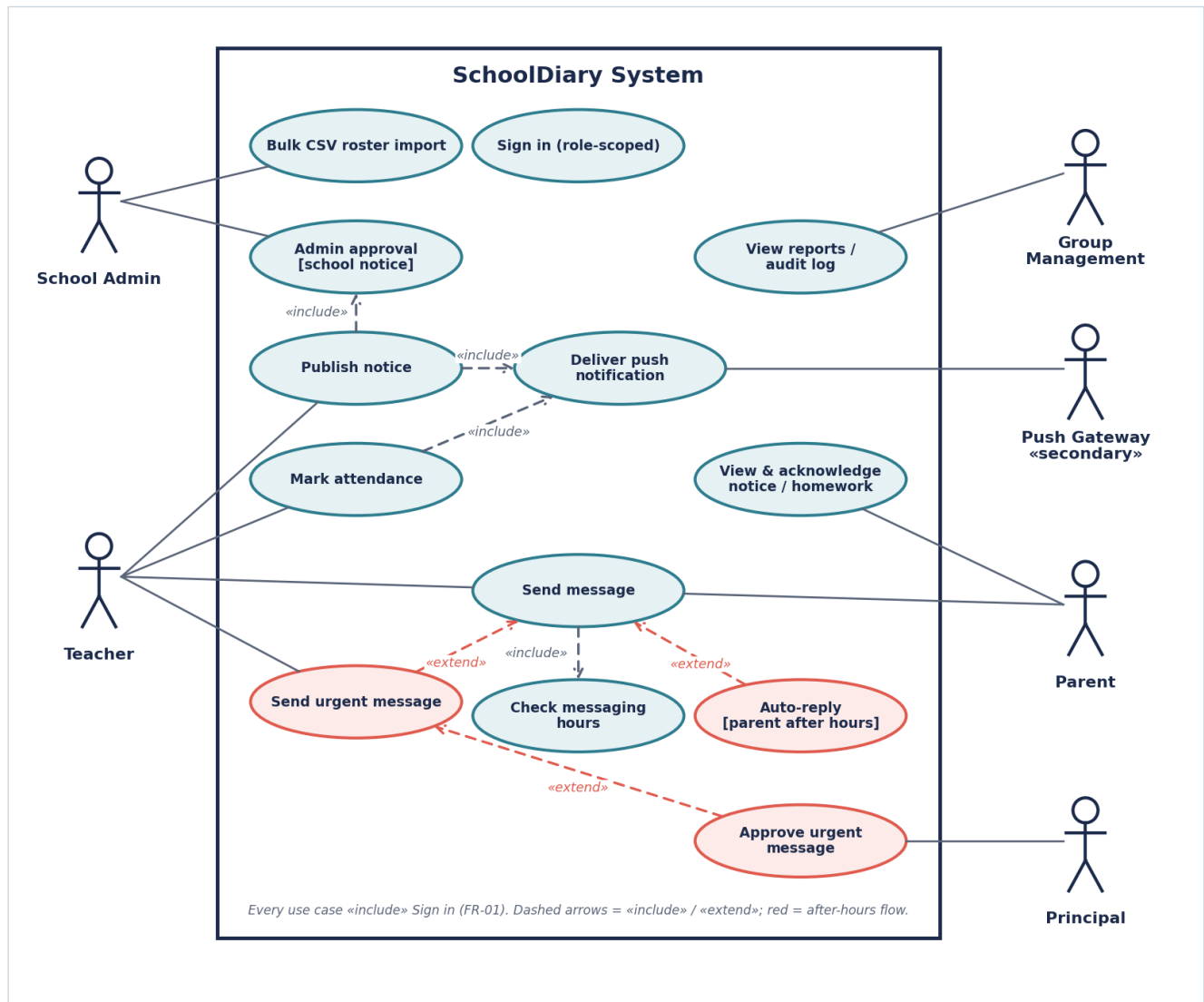


Figure 2.2: Use-Case Diagram — six actors, thirteen use cases, «include» / «extend» relationships

How to read this diagram

Actors sit outside the boundary: Parent, Teacher, School Admin, Principal, Group Management and the external Push Gateway (secondary system actor).

«include» — Send message always runs *Check messaging hours*; Publish notice always runs *Admin approval* (school scope) and *Deliver push notification*; every use case requires *Sign in* (FR-01).

«extend» (red, after-hours flow) — *Auto-reply* extends Send message for parents outside 07:00–19:59; *Send urgent message* extends Send message; *Approve urgent message* extends it for the Principal.

Traceability: FR-01, FR-02, FR-05, FR-07, FR-08, FR-09, FR-10 · Tests TC-01 to TC-10.

2.7.2 Class Diagram — Domain Model & Service Architecture

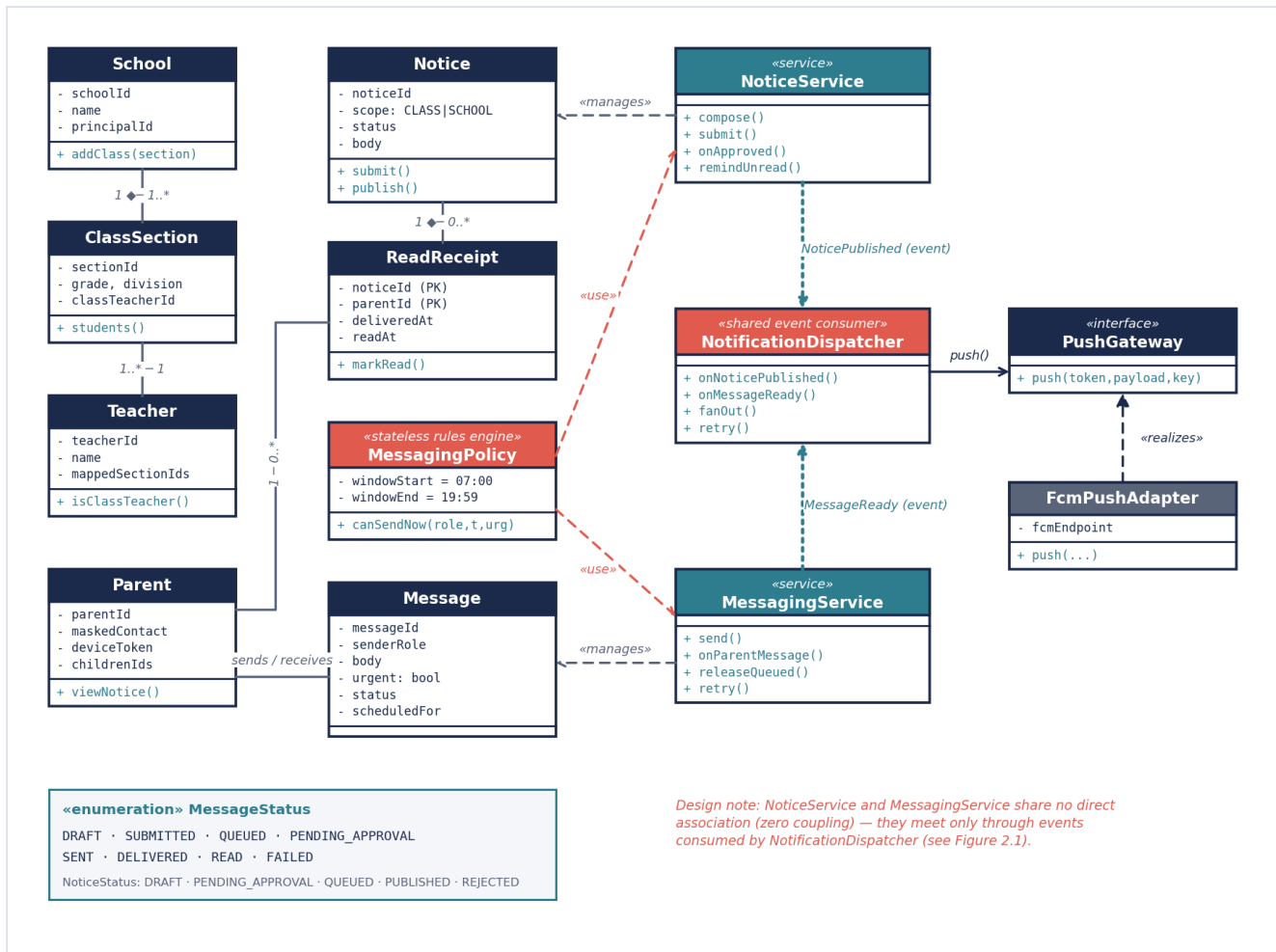


Figure 2.3: Class Diagram — domain aggregates, services, shared rules engine and push-gateway abstraction

How to read this diagram

Aggregate: School ♦ ClassSection; ClassSection maps to its class Teacher. ReadReceipt has a composite key (noticeId, parentId) so a parent can never be counted twice (FR-03).

Zero coupling: NoticeService and MessagingService share no method calls, interfaces or foreign keys; they only publish *NoticePublished* / *MessageReady* events consumed by NotificationDispatcher.

Single source of truth: both services consult the stateless MessagingPolicy (07:00–19:59 IST) so the window rule exists exactly once (FR-08).

Dependency inversion: the Dispatcher depends on the PushGateway interface; FcmPushAdapter is the only concrete class that knows about Google FCM.

Traceability: FR-02, FR-03, FR-07, FR-08 · NFR-02 (idempotency) · Figure 2.1.

2.7.3 Sequence Diagram — Send Notice & Track Receipts

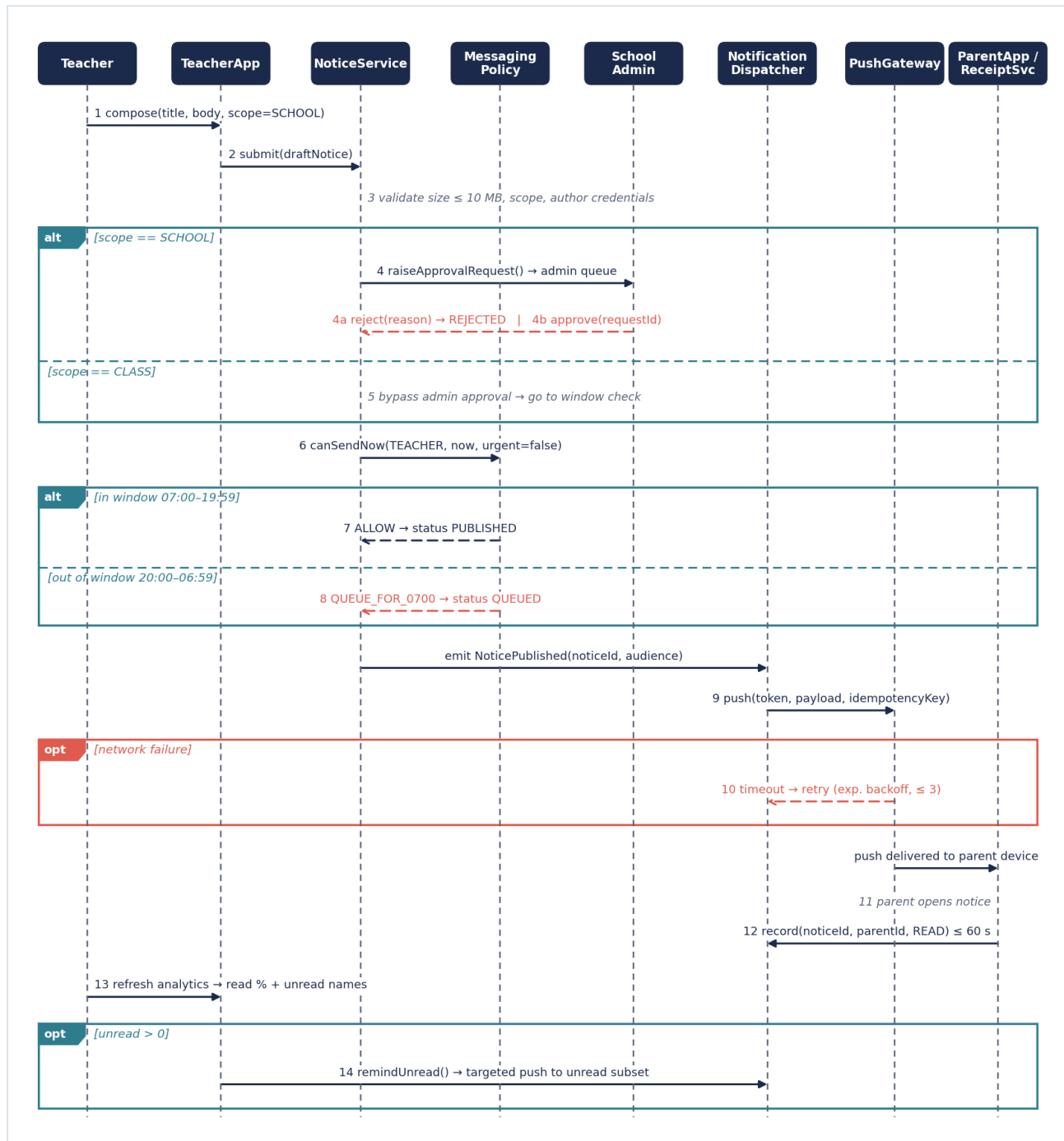


Figure 2.4: Sequence Diagram — school-level notice with approval, window check, fan-out, retry and read tracking

How to read this diagram

Steps 1–14 match the table in Section 2.3. The first **alt** frame separates school-level notices (admin approval, REJECTED or approved) from class-level notices (approval bypassed).

The second **alt** frame is the FR-08 check: in-window notices become PUBLISHED, out-of-window notices are QUEUED for 07:00. The **opt** frame shows ≤ 3 retries with exponential backoff (NFR-01, NFR-02); receipts persist within 60 s.

Traceability: FR-02, FR-03, FR-08 · TC-02, TC-03, TC-08.

2.7.4 Activity Diagram — Teacher Sends a Message

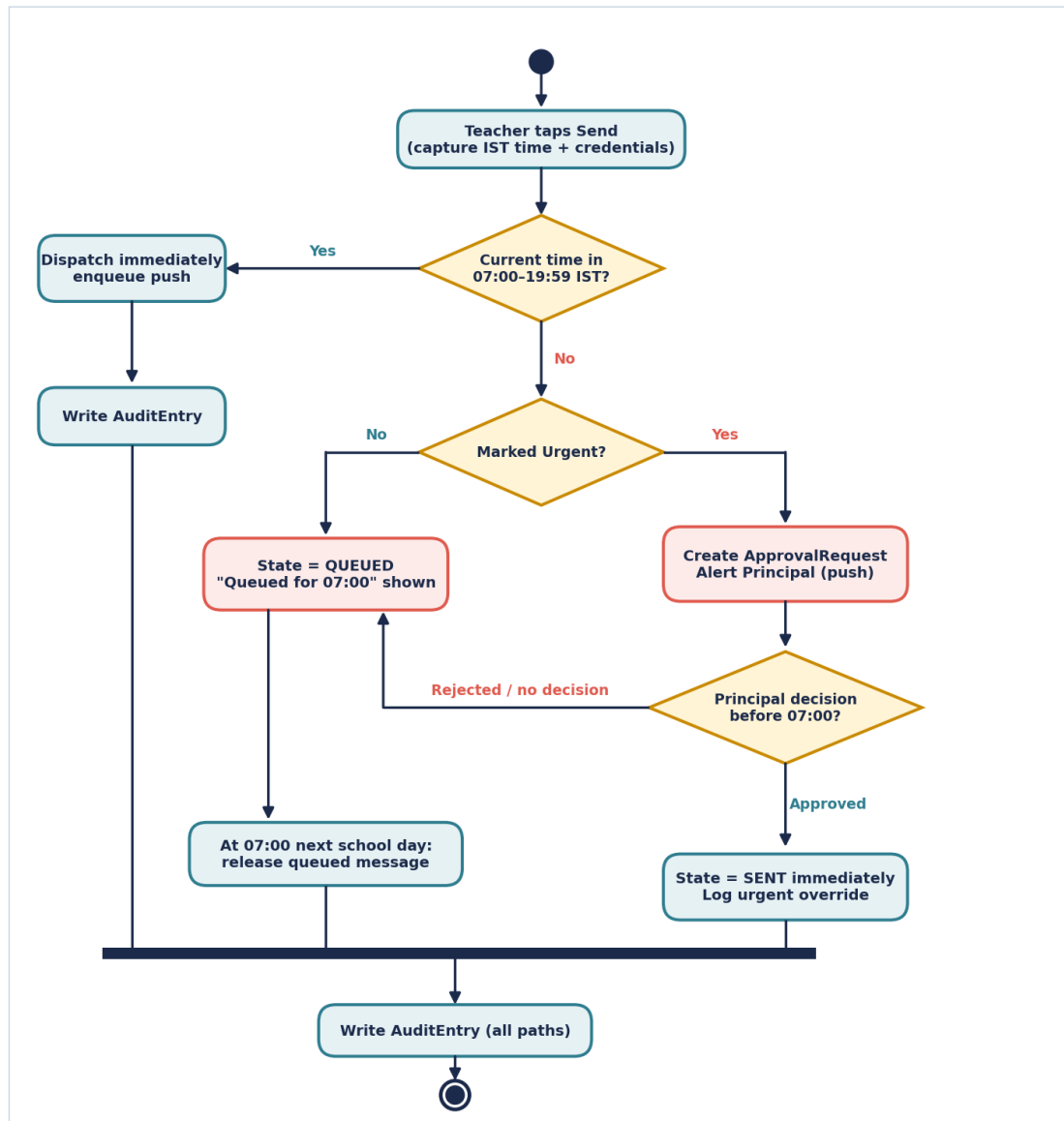


Figure 2.5: Activity Diagram — window check, urgency branch, Principal adjudication and 07:00 release

How to read this diagram

Decision 1 tests the 07:00–19:59 IST window; in-window messages are dispatched immediately.

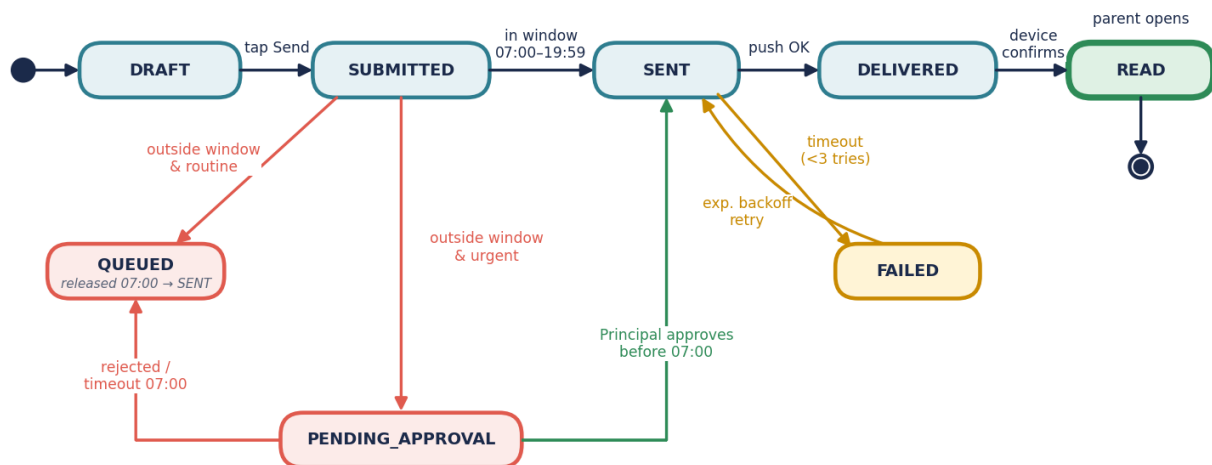
Decision 2 separates routine messages (QUEUED, shown as “Queued for 07:00”) from urgent ones, which raise an ApprovalRequest to the Principal.

Decision 3: approval before 07:00 sends immediately and logs an urgent override; rejection or no decision falls back to the normal 07:00 release (the DEF-28 scenario). Every path ends in an AuditEntry.

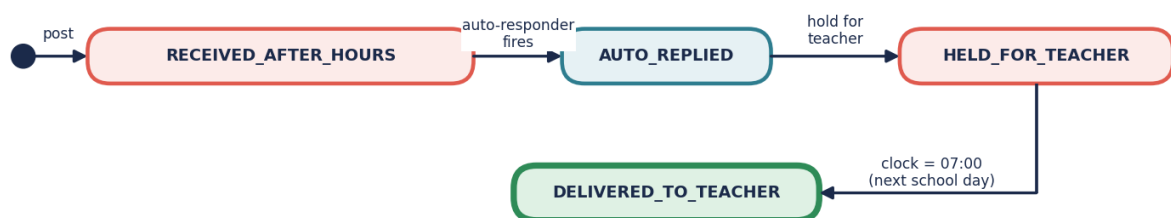
Traceability: FR-08, FR-10 · TC-BVA-01 to 09 · Decision-table rules R1–R5.

2.7.5 State Diagram — Message Lifecycle

A. Teacher-message lifecycle



B. Parent-message lifecycle (sent outside 07:00-19:59)



Guards: window = 07:00-19:59 IST (minute granularity). Parents are never blocked — only held for teacher review.

Retry policy: ≤ 3 attempts with exponential backoff; idempotency keys prevent duplicate delivery (NFR-02).

00:00 rollover rule: a message queued at 23:59 must create exactly ONE queue entry (regression guard for DEF-35).

Figure 2.6: State Diagram — teacher-message and parent-message lifecycles with guard conditions

How to read this diagram

Teacher messages: DRAFT → SUBMITTED, then SENT (in window), QUEUED (routine, out of window) or PENDING_APPROVAL (urgent, out of window). SENT may loop through FAILED (≤ 3 retries) before DELIVERED → READ.

Parent messages are never blocked: outside the window they receive an instant auto-reply, are held, and reach the teacher at 07:00 on the next school day.

Regression guards: DEF-15 (19:59 boundary), DEF-28 (rejection bypass) and DEF-35 (midnight double-queue) are all state-machine defects, which is why the Messaging module has the highest defect density in Section 6.7.

Traceability: FR-07, FR-08 · TC-BVA-10 to 18 · TC-EC-05 to 09.

SECTION 03

Project Plan & Critical Path Analysis

3.1 Work Breakdown Structure (WBS)

WBS Code	Work Package Name	Effort (pd)	Mapped Requirements	Key Engineering Deliverable
1.0	BUILD PHASE	51 pd	FR-01 to FR-10	Fully functional, verified application build
1.1 (A)	Notices & Circulars	8 pd	FR-02, FR-03	Class/school notice engine, approval workflow, read receipt analytics
1.2 (B)	Homework Module	10 pd	FR-04	Assignment posting, due date schedulers, parent acknowledgement tracking
1.3 (C)	Attendance Alerts	6 pd	FR-05	Roll-call absence trigger; automated 15-minute push notification pipeline
1.4 (D)	Fee Reminders	5 pd	FR-06	D-7 and D-1 automated reminder engine; strictly zero payment code
1.5 (E)	Messaging with Time Limits	12 pd	FR-07, FR-08	Two-way threads, 07:00-19:59 window engine, auto-responder, approval queue
1.6 (F)	Admin Console & Reports	10 pd	FR-01, FR-09, FR-10	Roster CSV bulk loader, mapping UI, audit log query engine, reports
2.0 (G)	VERIFY: Testing	12 pd	All Requirements	System test execution, boundary analysis, defect log, DRE certification
3.0 (H)	DEPLOY: Training	5 pd	Operational Rollout	Staff and administrative training across 2 pilot institutions; live launch
TOTAL	COMPLETE PILOT PROJECT	68 pd	—	Sum of 8 bottom-up work packages = 51 + 12 + 5 = 68 person-days

3.2 Network Diagram & Critical Path (CPM)

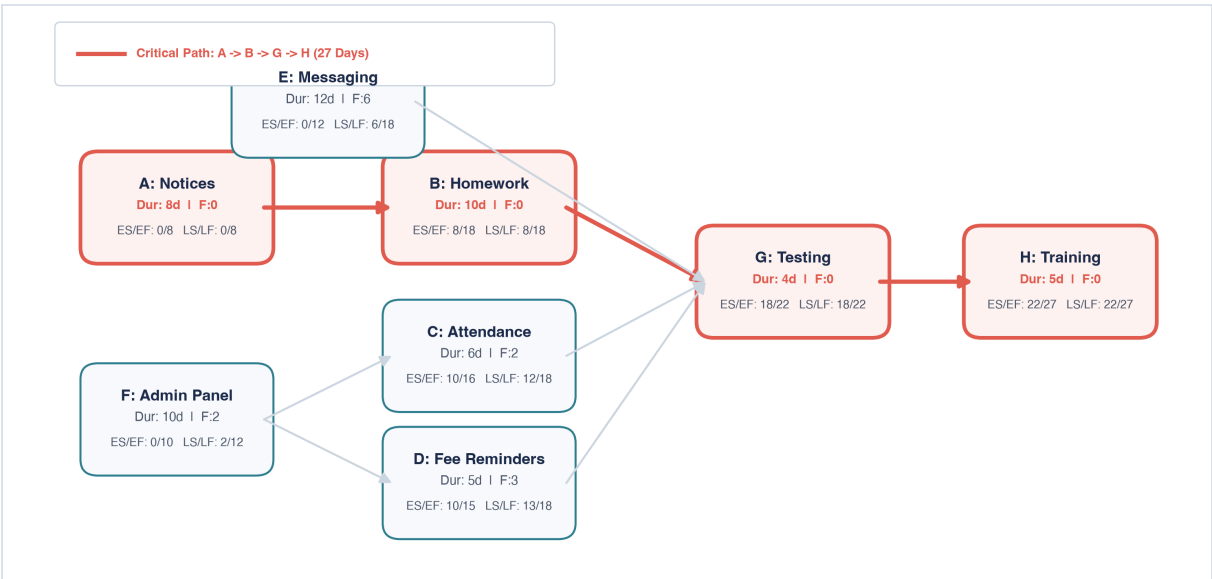


Figure 3.1: CPM Network Diagram — Critical Path A -> B -> G -> H (27 Working Days)

ID	Task Name	Duration	Predecessor s	ES	EF	LS	LF	Total Float	Criticality
A	Notices	8 days	—	0	8	0	8	0 days	CRITICAL PATH
B	Homework	10 days	A	8	18	8	18	0 days	CRITICAL PATH
C	Attendance Alerts	6 days	F	10	16	12	18	2 days	Non-Critical (Float 2d)
D	Fee Reminders	5 days	F	10	15	13	18	3 days	Non-Critical (Float 3d)
E	Messaging Hours	12 days	—	0	12	6	18	6 days	Non-Critical (Float 6d)
F	Admin Panel	10 days	—	0	10	2	12	2 days	Non-Critical (Float 2d)
G	System Testing	4 days*	B, C, D, E	18	22	18	22	0 days	CRITICAL PATH
H	Faculty Training	5 days	G	22	27	22	27	0 days	CRITICAL PATH

*Testing effort (12 person-days) is executed in parallel by the entire 3-person team: 12 ÷ 3 = 4 working days.

Critical Path Formula & Working: ES(G) = max(EF_B=18, EF_C=16, EF_D=15, EF_E=12) = 18 days.

Critical Path Sequence: Task A (8d) -> Task B (10d) -> Task G (4d) -> Task H (5d) = 27 working days.

Any delay in Tasks A, B, G, or H translates directly into a day-for-day slip of the pilot launch date.

3.3 Milestones & Gantt Schedule

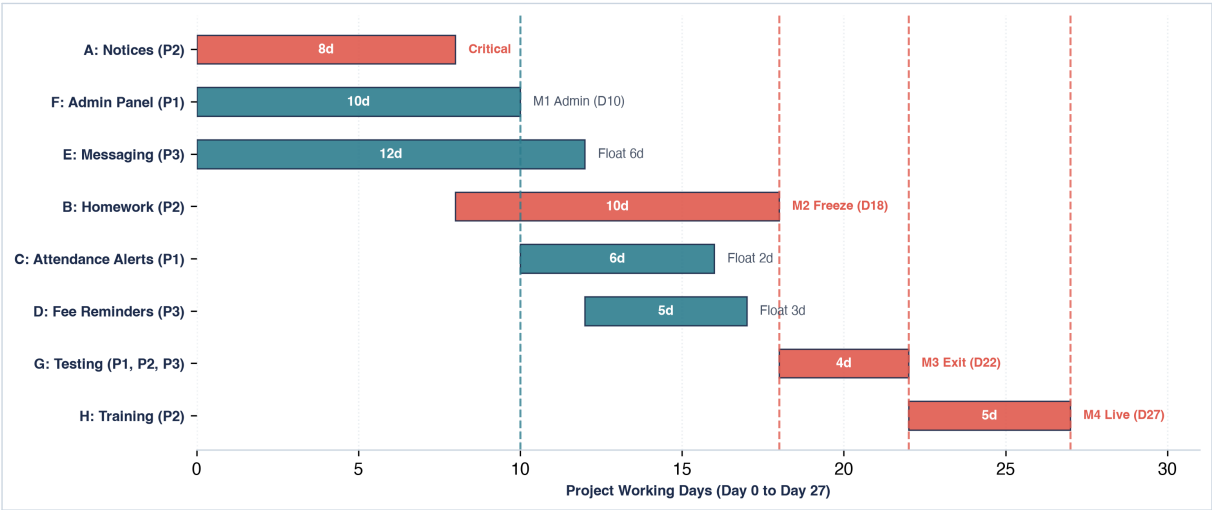


Figure 3.2: Project Gantt Schedule Across 27 Working Days with Milestone Gates

Milestone Gate	Target Day	Verifiable Exit Criteria & Evidence Required for Approval
M1: Admin Infrastructure Ready	Day 10	Admin console deployed; CSV roster loader verified; Tasks C and D unblocked.
M2: Feature Freeze Baseline	Day 18	All 6 functional modules built and unit-tested; test data loaded; feature code frozen.
M3: Quality Exit & Go/No-Go	Day 22	Testing phase complete; zero open critical or high defects; DRE confirmed; sign-off.
M4: Pilot Operational Go-Live	Day 27	Faculty training concluded at 2 pilot schools; system deployed to 2,300 pilot parents.

3.4 Resource Allocation & Schedule Analysis

Engineer Profile	Days 0–10	Days 10–16	Days 16–18	Days 18–22	Days 22–27
P1: Lead Developer	Task F: Admin Panel	Task C: Attendance	Prepare test suites	Task G: Testing (All)	Post-launch pilot defect support

Engineer Profile	Days 0–10	Days 10–16	Days 16–18	Days 18–22	Days 22–27
P2: Frontend Developer	Task A: Notices (0–8d)	Task B: Homework	Task B: Homework	Task G: Testing (All)	Task H: Lead Trainer (2 schools)
P3: Backend / QA Dev	Task E: Messaging (0–12d)	Task D: Fee Reminders	Task D finish & test data	Task G: Testing (All)	Post-launch pilot defect support

Strategic Schedule Rationale — Why 27 Days and Not 23 Days?

The theoretical ideal duration equals total effort divided by headcount: $68 \div 3 = 22.67 \approx 23$ working days. However, scheduling a project at 23 days is mathematically impossible due to physical dependencies:

- Serial Dependency Chain:** Task B (Homework) reuses publishing infrastructure from Task A (Notices). Hence, Task B cannot conclude before Day 18 ($8 + 10 = 18$ d).
- Testing Feature Freeze Precondition:** Formal system and UAT verification (Task G) requires all features to be integrated, making Day 18 the earliest possible start.
- Single-Trainer Constraint:** Institutional training (Task H) across 2 schools is conducted by Developer P2, requiring 5 contiguous working days that cannot be compressed.

Schedule Stretch: 27 days – 23 days = 4 working days of schedule stretch, yielding an 84% resource utilisation rate ($68 \div 81$ pd).

SECTION 04

Estimation Sheet & Mathematical Proofs

4.1 Inputs & Stated Assumptions

ID	Category	Stated Assumption & Engineering Basis
A1	Demographic Distribution	Parent population is distributed evenly across 12 schools: $14,000 \div 12 \approx 1,167$ parents per institution.
A2	Faculty Distribution	Teacher staffing is directly proportional to parent count: pilot teachers = $650 \times (2,300 \div 14,000) \approx 107$ teachers.
A3	Effort Productivity	Each person-day represents 1 productive working day for 1 engineer; estimates include local code reviews.
A4	Parallelism Bounds	Ideal duration assumes perfect, unconstrained parallel execution across 3 developers.
A5	CPM Network Rigor	Scheduled duration is strictly derived from the Critical Path Method forward/backward pass (Section 3).
A6	Calendar Framework	All durations represent elapsed working days (Mon–Sat); project start is established as Day 0.
A7	Active Parent Definition	A parent is classified as Monthly Active if they open ≥ 1 circular, homework, or thread within the calendar month.

4.2 Step-by-Step Calculations & Formulas

#	Metric	Mathematical Formula	Exact Step-by-Step Working	Computed Result
1	Pilot Share of Parents	Pilot Parents $\div$ Total Parents	$2,300 \div 14,000 = 0.1642857\dots$	16.43% ($\approx 16.4\%$)
2	Group Parents / School	Total Parents $\div$ Schools	$14,000 \div 12 = 1,166.67\dots$	$\approx 1,167$ parents
3	Pilot Parents / School	Pilot Parents $\div$ Pilot Schools	$2,300 \div 2 = 1,150.0$	1,150 parents
4	Pilot Faculty Allocation	Total Teachers $\times$ Pilot Share	$650 \times 0.1642857\dots = 106.78\dots$	≈ 107 teachers
5	Total Bottom-Up Effort	Sum of 8 Work Packages	$8 + 10 + 6 + 5 + 12 + 10 + 12 + 5 = 68$	68 person-days
6	Ideal Duration	Total Effort $\div$ Headcount	$68 \div 3 = 22.666\dots$	$22.67 \approx 23$ working days
7	Scheduled Duration	Critical Path A + B + G + H	$8 + 10 + 4 + 5 = 27$	27 working days
8	Available Capacity	Headcount $\times$ Scheduled Duration	$3 \times 27 = 81$	81 person-days
9	Resource Utilisation	Total Effort $\div$ Available Capacity	$68 \div 81 = 0.839506\dots$	$83.95\% \approx 84\%$
10	Schedule Stretch	Scheduled Duration – Ideal Duration	$27 - 22.67 = 4.33\dots$	4 working days
11	Pilot Active Target	$90\% \times$ Pilot Parents	$0.90 \times 2,300 = 2,070$	2,070 parents
12	Full Rollout Active Target	$90\% \times$ Total Parents	$0.90 \times 14,000 = 12,600$	12,600 parents
13	Late Messaging Target	Direct SLA Objective	Zero messages delivered after 20:00	0 messages

4.3 Confidence Level & Why the Result is an Estimate, Not a Promise

Estimation Metric	Baseline Point Estimate	Optimistic Bound (-20%)	Pessimistic Bound (+20%)
Engineering Effort	68.0 person-days	$68.0 \times 0.80 = 54.4$ person-days	$68.0 \times 1.20 = 81.6$ person-days
Ideal Duration (3 Engineers)	22.7 working days	$54.4 \div 3 = 18.1$ working days	$81.6 \div 3 = 27.2$ working days
Scheduled Duration (CPM)	27.0 working days	Fast-tracked: 23 working days	Buffer-adjusted: 32 working days

Professional Justification: An Estimate is a Probability Distribution, Not a Contractual Promise:

The figures of 68 person-days, 23 ideal days, and 27 scheduled days are professional engineering estimates developed during the requirements elicitation stage prior to architectural design. Consequently, an honest industry confidence band of $\pm 20\%$ applies (54.4 to 81.6 person-days).

- Assumption Sensitivity:** The estimate presumes immediate CSV roster availability from pilot schools, zero engineer attrition, and a maximum of 3 push retry cycles. If any assumption fails, effort variance increases.
- Complexity Variance:** Task E (Messaging Hours) represents the single largest package (12 pd). While backed by 6 days of total float, its state-machine edge cases represent the highest variance risk.
- Commitment vs Forecast:** A promise represents a rigid commitment regardless of newly discovered scope; an estimate is a data-driven forecast refined continuously at each milestone gate (M1 to M4).

SECTION 05

Scope Management & Change Control

5.1 Pilot Scope Statement

Pilot Scope Boundary Specification: Deliver a fully operational communication suite to 2 pilot schools, encompassing 2,300 parents and ~107 faculty members. Deliverables comprise an Android native application, web admin portal, and all 8 core functional modules (FR-01 to FR-10) within 68 person-days and 27 working days. Key success targets: 90% monthly active parents (2,070 parents) and 0 late teacher messages after 20:00.

5.2 Scope Boundary & Impact on Time, Cost & Quality

Scope Work Item	Lifecycle Boundary	Impact on Schedule (Time)	Impact on Budget (Cost)	Impact on Engineering Quality
Notices & Receipts (FR-02, 03)	IN SCOPE (Pilot)	Task A (8d): On critical path; slip delays go-live directly	8 person-days allocated	Eliminates missed circulars; captures read verification
Homework Module (FR-04)	IN SCOPE (Pilot)	Task B (10d): On critical path; drives Day 18 feature freeze	10 person-days allocated	Standardizes assignments; reuses notice delivery pipeline
Attendance Alerts (FR-05)	IN SCOPE (Pilot)	Task C (6d): Non-critical (2d float absorbs small slips)	6 person-days allocated	Delivers 15-min safety alert; enforces roster integrity
Fee Reminders (FR-06)	IN SCOPE (Pilot)	Task D (5d): Non-critical (3d float absorbs small slips)	5 person-days allocated	Automates collection chasing without payment compliance burden
Messaging Hours (FR-07, 08)	IN SCOPE (Pilot)	Task E (12d): Largest item; 6d float isolates critical path	12 person-days allocated	Protects faculty work-life balance; stops night message spam
Admin Panel (FR-01, 09, 10)	IN SCOPE (Pilot)	Task F (10d): Non-critical (2d float); hits M1 on Day 10	10 person-days allocated	Enforces strict tenant scoping; central audit logging
Testing & Training (G, H)	IN SCOPE (Pilot)	12 pd testing (4d) + 5d training: Both on critical path	17 person-days allocated	Testing guarantees 94.4% DRE; training drives 90% adoption
10 Rollout Schools	Later Release	Zero pilot schedule impact; planned for Phase 2	Funded under Phase 2 budget	Pilot defect data hardens codebase before multi-tenant scale
C-01 Multilingual UI	Later Release	Deferred; adding i18n would inflate critical path by 5d	Requires external translation	Improves vernacular reach; deferred to safeguard 27d launch
C-02 SMS Fallback Gateway	Later Release	Deferred; telecom gateway integration adds 4d	Per-SMS carrier charges	Crucial contingency for offline parents; prioritised for rollout
C-03 PTM Slot Booking	Later Release	Deferred; calendar conflict engine requires 8d	Additional dev budget	High user value; non-essential for initial communication problem

Scope Work Item	Lifecycle Boundary	Impact on Schedule (Time)	Impact on Budget (Cost)	Impact on Engineering Quality
Native iOS Client	Later Release	Deferred; dual-platform mobile build doubles UI effort	Requires iOS dev & test kits	Android covers 96% of pilot demographic; iOS deferred
Online Fee Payment	OUT OF SCOPE	Excluded; PCI-DSS compliance & gateway testing adds 15d	Merchant fees & liability	Eliminates financial security, fraud, and audit liabilities
Real-Time Video Calls	OUT OF SCOPE	Excluded; WebRTC infrastructure adds 20d	High server streaming costs	Unusable on 2GB RAM budget phones and constrained data plans
Student User Accounts	OUT OF SCOPE	Excluded; child identity management adds 8d	Additional privacy audits	Eliminates child COPPA/DPDP regulatory compliance risks
WhatsApp API Bridge	OUT OF SCOPE	Excluded; Meta BSP approval timeline is unpredictable	Per-conversation Meta fees	Defeats project objective by enabling out-of-hours message leaks

5.3 Scope-Change Control Procedure

Process Step	Governing Action & Evaluation Protocol	Accountable Role	Enforcement SLA
1. Request Submission	Formal Change Request (CR) logged with business justification and technical scope description.	Any Stakeholder	Continuous logging
2. Impact Assessment	Evaluate CPM impact. If critical task affected -> project finish extends. If non-critical -> evaluate float. Recompute effort and cost.	Project Manager	Within 2 working days
3. Governance Review	Management Sponsor evaluates CR. Decisions: Approve, Reject, or Defer to Phase 2 backlog. Changes >81.6 pd require re-estimation.	Management Sponsor	Within 3 working days
4. Baseline Re-alignment	If approved, formally re-baseline SRS, CPM schedule, Estimation Sheet, and Test Plan under new version number.	Project Manager	Prior to task start
5. Stakeholder Notice	Formal notification of schedule adjustments disseminated to faculty, admins, and engineering team.	Project Manager	Immediate dispatch

Simulated Scope Creep Rejections During Pilot Formulation:

- *Payment Processing Integration*: Requested by school accounts; rejected due to PCI-DSS liability and 15-day critical path delay.
- *WhatsApp Notification Mirror*: Requested by faculty; rejected because external messaging bypasses the 20:00 cutoff and audit trail.
- *Student Direct Accounts*: Requested by senior teachers; rejected to eliminate child privacy and consent compliance exposure.

SECTION 06

Test Plan & Empirical Evidence

IEEE 829-lite Test Specification · Identifier: TP-SD-01

6.1 Test Strategy, Levels & Environmental Controls

Test Strategy Dimension	Operational Specification & Environmental Governance
Entry Criteria	Milestone M2 reached (Day 18); feature freeze enacted; synthetic rosters loaded; rules oracle confirmed.
Exit Criteria	100% of planned tests executed; zero open critical or high defects at Milestone M3 (Day 22); DRE certified ≥90%.
Multi-Level Execution	Unit (BVA, Equivalence Partitioning, Decision Table); Integration (API, gateway retry); System (TC-01–10); UAT.
Test Bed Environment	Synthetic roster data only; 2 test Android 8 handsets (2 GB RAM) + modern Android device; web admin console; gateway clock mock.
Governance Roles	P3 (Developer/QA) leads test execution; P1/P2 resolve defects; School Admins & Principals participate in UAT sign-off.

6.2 System Test Cases TC-01 to TC-10

Case ID	Target FR	Execution Procedure & Operational Scenario	Expected Result & Verification Oracle
TC-01	FR-01	Log in as Parent, Teacher, Admin, Mgmt. Attempt cross-tenant and cross-child access.	Access granted to authorized scope; 403 Forbidden on cross-scope requests.
TC-02	FR-02	Publish class circular (PDF). Submit school-level circular. Admin approves.	Class notice posts immediately; school circular stays Pending until admin approval.
TC-03	FR-03	Dispatch circular to 10 test parents. Open as 6 parents. Inspect dashboard.	Dashboard displays exactly 60% read; unread roster lists 4 remaining parents by name.
TC-04	FR-04	Post Math homework with due date. Open Parent client and tap Acknowledge.	Homework renders with due date; parent acknowledgement persists in faculty view.
TC-05	FR-05	Submit roll call marking Student A absent at 09:00. Time push delivery.	Alert delivered to Student A parent device by 09:15 (<=15 min); zero alert for present pupils.
TC-06	FR-06	Seed fee due date D. Advance mock system clock to D–7 and D–1.	Automated notifications trigger at D–7 and D–1; interface contains zero payment UI.
TC-07	FR-07	Parent messages teacher at 10:30. Teacher replies at 10:40.	Single synchronized thread per child; mutual contact phone numbers masked as XXXX-XXXX.
TC-08	FR-08	Execute full battery of BVA (18 cases) and Decision Table (13 rules).	All tests match oracle: Immediate send, queue for 07:00, approval, or parent hold.
TC-09	FR-09	Bulk upload CSV of 100 pupils with 1 corrupt row. Map faculty to sections.	99 pupils committed; 1 error flagged with exact row and column; mappings update.
TC-10	FR-10	Search communication audit log. Generate Active Parents and Late Message reports.	Audit query returns exact match; reports show accurate numerator, denominator, and %.

6.3 Equivalence Classes (Message Time & Input Partitions)

Partition ID	Input Domain / Condition	Validity	Expected System Behavior	Representative Value
EC-T1	07:00 <= t <= 19:59 (Teacher Sending Window)	Valid	In-window: Immediate gateway dispatch	13:30 IST
EC-T2	00:00 <= t < 07:00 (Pre-Window Early Morning)	Valid	Out-of-window: Enqueued for 07:00 release	03:15 IST
EC-T3	20:00 <= t <= 23:59 (Post-Window Night)	Valid	Out-of-window: Enqueued for next school day 07:00	21:45 IST
EC-T4	Malformed Time Input (hour > 23, minute > 59)	Invalid	Input validation error: Rejected at API boundary	25:10 IST
TC-EC-05	Parent Sender inside 07:00–19:59 IST	Valid	Immediate delivery to faculty thread	13:30 IST
TC-EC-06	Parent Sender outside 07:00–19:59 IST	Valid	Held for teacher at 07:00 + instant auto-reply	21:45 IST
TC-EC-07	Teacher Urgent Message, Out of Window, Pending	Valid	ApprovalRequest enqueued for Principal review	21:45 IST
TC-EC-08	Teacher Urgent Message, Out of Window, Approved	Valid	Immediate emergency dispatch override	21:45 IST
TC-EC-09	Teacher Urgent Message, Out of Window, Rejected	Valid	Queued for regular morning dispatch at 07:00	21:45 IST
TC-EC-10	Class Notice: In-Window vs Out-of-Window	Valid	Immediate publish (in-window) vs Queued for 07:00	10:30 / 21:45 IST
TC-EC-11	School Notice Submitted by Teacher	Valid	Routes to School Admin queue; Pending status	10:30 IST

6.4 Boundary Value Analysis Around 8 p.m. & 7 a.m. (18 Test Cases)

Test ID	Sender Role	Exact Timestamp	Expected System Action & Test Outcome Significance
TC-BVA-01	Teacher (Routine)	06:59 IST	Blocked; enqueued for 07:00. Faculty client displays "Queued for 07:00".
TC-BVA-02	Teacher (Routine)	07:00 IST	Allowed; immediate dispatch (Opening boundary minute of sending window).
TC-BVA-03	Teacher (Routine)	07:01 IST	Allowed; immediate dispatch (Inside window).
TC-BVA-04	Teacher (Routine)	19:58 IST	Allowed; immediate dispatch (Inside window).
TC-BVA-05	Teacher (Routine)	19:59 IST	Allowed; immediate dispatch. ■ Caught DEF-15 (off-by-one bug where t<=19:58 was coded).
TC-BVA-06	Teacher (Routine)	20:00 IST	Blocked; enqueued for next school day 07:00 (Closing boundary minute).
TC-BVA-07	Teacher (Routine)	20:01 IST	Blocked; enqueued for next school day 07:00 (After-hours window).
TC-BVA-08	Teacher (Routine)	23:59 IST	Blocked; enqueued for next school day 07:00 (Pre-midnight boundary).

Test ID	Sender Role	Exact Timestamp	Expected System Action & Test Outcome Significance
TC-BVA-09	Teacher (Routine)	00:00 IST (Day D+1)	Blocked; enqueued for same Day D+1 07:00. Exactly 1 queue entry. ■ Caught DEF-35.
TC-BVA-10	Parent Client	06:59 IST	Held for faculty at 07:00. Instant auto-reply dispatched to parent.
TC-BVA-11	Parent Client	07:00 IST	Allowed; delivered immediately to faculty message inbox; zero auto-reply.
TC-BVA-12	Parent Client	07:01 IST	Allowed; delivered immediately to faculty message inbox.
TC-BVA-13	Parent Client	19:58 IST	Allowed; delivered immediately to faculty message inbox.
TC-BVA-14	Parent Client	19:59 IST	Allowed; delivered immediately to faculty message inbox.
TC-BVA-15	Parent Client	20:00 IST	Held until next school day 07:00. Instant auto-reply dispatched to parent.
TC-BVA-16	Parent Client	20:01 IST	Held until next school day 07:00. Instant auto-reply dispatched to parent.
TC-BVA-17	Parent Client	23:59 IST	Held until next school day 07:00. Instant auto-reply dispatched to parent.
TC-BVA-18	Parent Client	00:00 IST (Day D+1)	Held until Day D+1 07:00 once only. Instant auto-reply dispatched to parent.

6.5 Decision Table for Messaging-Hours & Approval Rules (13 Rules)

Condition Matrix: C1=Sender Role (T/P/A), C2=In Window (Y/N), C3=Urgent Flag (Y/N/—), C4=Principal Approved (Y/N/Pending/—), C5=Item Scope (Msg/Class/School), C6=Decision Final (Y/N)

Rule	C1: Role	C2: Window	C3: Urgent	C4: Approved	C5: Scope	System Action Executed by Policy Engine
R1	Teacher	Yes (07–20)	—	—	Message	A1: Send immediately via NotificationDispatcher
R2	Teacher	No (20–07)	No	—	Message	A2: Enqueue for automated release at 07:00 next school day
R3	Teacher	No (20–07)	Yes	Pending	Message	A3: Create ApprovalRequest; alert Principal; hold delivery
R4	Teacher	No (20–07)	Yes	Approved	Message	A1: Send immediately (Emergency Principal override dispatched)
R5	Teacher	No (20–07)	Yes	Rejected/Ti meout	Message	A2: Enqueue for 07:00 next school day. ■ Caught DEF-28.
R6	Parent	Yes (07–20)	—	—	Message	A1: Deliver immediately to teacher inbox
R7	Parent	No (20–07)	—	—	Message	A4: Deliver auto-reply to parent; hold message for teacher until 07:00
R8	Teacher	Yes (07–20)	—	—	Class Notice	A5: Publish immediately to assigned classroom parent roster
R9	Teacher	No (20–07)	—	—	Class Notice	A2: Enqueue class notice for automated publishing at 07:00
R10	Teacher	—	—	—	School Notice	A6: Route to School Admin approval queue; mark PENDING_APPROVAL
R11	Admin	Yes (07–20)	—	—	School Notice	A5: Publish immediately across entire school parent body
R12	Admin	No (20–07)	—	—	School Notice	A2: Enqueue school notice for 07:00 morning release

Rule	C1: Role	C2: Window	C3: Urgent	C4: Approved	C5: Scope	System Action Executed by Policy Engine
R13	Admin	—	—	—	School Notice	A7: Return notice to draft; alert sender with rejection reason

6.6 Empirical Test Log & Defect Distribution

Testing Level	Planned Cases Run	Failed Cases (Defects)	Passed Cases	Execution Pass Rate
Unit Testing (BVA, EC, Decision Table)	90 cases	11 defects	79 cases	87.78%
Integration Testing (API & Gateway Mocks)	30 cases	7 defects	23 cases	76.67%
System Testing (TC-01 to TC-10 Scenarios)	40 cases	6 defects	34 cases	85.00%
User Acceptance Testing (UAT - Faculty & Parents)	15 cases	3 defects	12 cases	80.00%
TOTAL PRE-RELEASE EXECUTION	175 cases	27 defects*	148 cases	84.57%

*Note: The 27 test-phase defects combine with 7 review defects (4 requirements review + 3 design review) to yield 34 pre-release defects.

Engineering Lifecycle Phase	Total Defects	Critical Severity	High Severity	Medium Severity	Low Severity
Requirements Formal Inspection	4 defects	0	1	2	1
Architectural Design Review	3 defects	0	1	1	1
Unit Testing Execution	11 defects	0	1	5	5
Integration Testing Execution	7 defects	1	2	3	1
System Testing Execution	6 defects	0	1	3	2
User Acceptance Testing (UAT)	3 defects	0	0	1	2
Post-Release Pilot (Weeks 1–2)	2 defects	0	1	1	0
CUMULATIVE DEFECT INVENTORY	36 defects	1	7	16	12

Defect ID	Lifecycle Phase	Severity	Root Cause & Empirical Failure Description	Verification Oracle
DEF-15	Unit Test	High	Off-by-one error: Sending window coded as t <= 19:58 instead of 19:59.	Caught by TC-BVA-05 (19:59 boundary)
DEF-21	Integration	Critical	Authorization bypass: ClassSection mapping missing in message service; cross-class leak.	Caught by role security test harness
DEF-28	System Test	High	Principal rejection bypassed: Message dispatched immediately at 21:20 instead of 07:00.	Caught by TC-DT-R5 (decision table)

Defect ID	Lifecycle Phase	Severity	Root Cause & Empirical Failure Description	Verification Oracle
DEF-35	Post-Release	High	Midnight rollover double queue: Message sent at 23:59 re-queued at 00:00 -> dual delivery.	Caught in Pilot W1; patched immediately

6.7 Defect Removal Efficiency (DRE) & Defect Density

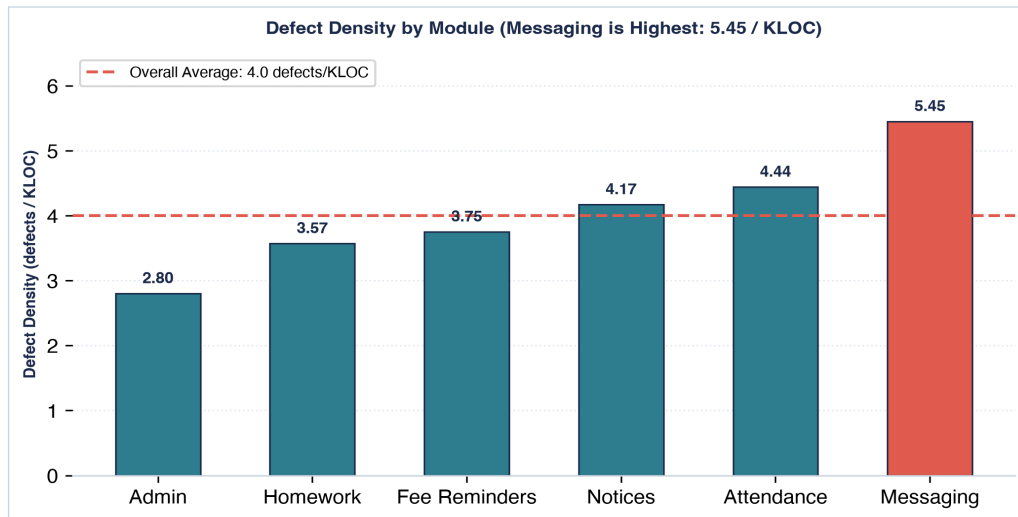


Figure 6.1: Defect Density per Module Against the 4.0 Defects/KLOC Benchmark

Defect Removal Efficiency (DRE) Mathematical Proof:

Formula: $DRE = E \div (E + D)$ where E = pre-release defects found, D = post-release defects in pilot.

- Pre-Release Defects (E) = 4 (Req) + 3 (Design) + 11 (Unit) + 7 (Integ) + 6 (System) + 3 (UAT) = **34 defects**
- Post-Release Pilot Defects (D) = **2 defects** (DEF-35 rollover bug + DEF-36 read % UI anomaly)

$DRE = 34 \div (34 + 2) = 34 \div 36 = 0.94444... = 94.4\%$ (Target: $\geq 90\%$ EXCEEDED)

Test-Phase Only DRE: Considering only executable test stages ($E_{test} = 27$): $27 \div (27 + 2) = 27 \div 29 = 93.1\%$.

Module Name	Size (KLOC)	Defect Count	Defect Density (Defects/KLOC)	Comparative Quality Interpretation
Notices & Circulars	1.2 KLOC	5 defects	4.17 defects / KLOC	Close to group average; approval logic contributed 2 bugs
Homework Module	1.4 KLOC	5 defects	3.57 defects / KLOC	Below average; reuse of notice publishing pipeline mitigated defects
Attendance Alerts	0.9 KLOC	4 defects	4.44 defects / KLOC	Slightly elevated due to roll-call re-save duplicate bug
Fee Reminders	0.8 KLOC	3 defects	3.75 defects / KLOC	Below average; simple cron-based notification scheduling
Messaging Hours Engine	2.2 KLOC	12 defects	5.45 defects / KLOC	■ Highest density (36% above average); contains 1/3 of all defects; high risk
Admin Web Console	2.5 KLOC	7 defects	2.80 defects / KLOC	Lowest density; standard CRUD screens with low architectural complexity
SYSTEM TOTAL / BENCHMARK	9.0 KLOC	36 defects	4.00 defects / KLOC	Overall baseline: 36 defects $\div$ 9.0 KLOC = 4.0 defects / KLOC

SECTION 07

Risk Management & Issue Log

7.1 Scoring Methodology & Probability × Impact Heatmap

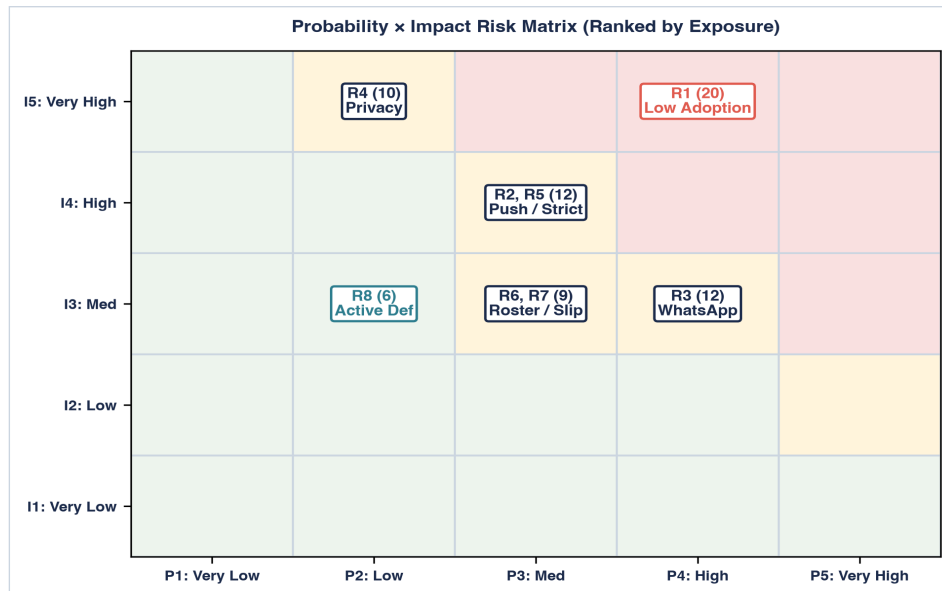


Figure 7.1: Probability × Impact Risk Matrix Heatmap with 8 Exposure-Ranked Risks

Risks are scored using 5-point ordinal scales: Probability (P1=Very Low to P5=Very High) and Impact (I1=Negligible to I5=Catastrophic). **Exposure Score = $P \times I$** (Range 1 to 25). Bands: High Exposure (≥ 15), Medium Exposure (9–14), Low Exposure (≤ 8). Ties broken by higher impact, then breadth of affected users.

7.2 Risk Register (8 Exposure-Ranked Risks)

Rank	ID	Risk Description	P	I	Exposure	Band	Mitigation Strategy	Accountable Owner
1	R1	Low parent adoption (<90% monthly active target)	4	5	20	HIGH	Mitigate: Teacher-led onboarding drive; mirror circulars; SMS fallback	Project Manager & Management Sponsor
2	R2	Push delivery failure on budget phones / battery savers	3	4	12	MED	Mitigate: Test on 2GB RAM phones; in-app inbox; idempotent retries	Lead Developer (P1)
3	R5	Messaging rule too strict; genuine emergencies delayed	3	4	12	MED	Mitigate: Principal urgent override; school phone in auto-reply; audit	Developer/QA (P3) with Principals
4	R3	Faculty continue running unofficial WhatsApp groups	4	3	12	MED	Mitigate: Principal policy circular; publish after-8 p.m. compliance reports	Pilot School Principals
5	R4	Tenant privacy breach: parent sees other child data	2	5	10	MED	Avoid: Service-layer role checks; masked contacts; penetration tests	Lead Developer (P1)
6	R6	Roster CSV data import corruption and invalid phones	3	3	9	MED	Transfer: School admins sign off rosters; admin dry-run syntax check	School Administrative Staff

Rank	ID	Risk Description	P	I	Exposure	Band	Mitigation Strategy	Accountable Owner
7	R7	Schedule slip in Messaging module (12 pd budget)	3	3	9	MED	Accept: Task E has 6d float; reassign P1 to Task D if E slips >1d	Project Manager
8	R8	Management disputes "Monthly Active Parent" definition	2	3	6	LOW	Avoid: Formal written consensus on definition prior to deployment	Project Manager

7.3 RMMM Plans for Top 3 Risks

Risk 1: Low Parent Adoption (Exposure Score: 20 — HIGH BAND)

RMMM Dimension	Operational Protocol & Strategic Action Plan
Risk Mitigation	Conduct in-person onboarding drives during parent orientations at both pilot schools; distribute QR download sheets in paper diaries; mandate that official circulars are published exclusively on SchoolDiary; maintain school-office walk-in support desks.
Risk Monitoring	Weekly inspection of Monthly Active Parent compliance reports (FR-10).
Trigger Metric	Active parent adoption drops below 80% (1,840 parents) by end of Week 3, OR 24-hour circular read rate falls below 75%.
Contingency Management	Immediately fast-track C-02 SMS Fallback integration; deploy classroom parent coordinators to assist non-registered families; steering committee convenes to assess adoption barriers prior to rollout expansion.
Accountable Owner	Project Manager with Group Management Sponsor

Risk 2: Push Delivery Failure / Offline Budget Handsets (Exposure Score: 12 — MEDIUM)

RMMM Dimension	Operational Protocol & Strategic Action Plan
Risk Mitigation	Rigorous physical testing on Android 8 devices with 2 GB RAM and active OEM aggressive battery savers; implement idempotent delivery queues (NFR-02); provide prominent in-app notification inbox ensuring notices are visible even when system push fails.
Risk Monitoring	Daily gateway telemetry monitoring; measure 5-minute delivery success against the 99% SLA (NFR-01).
Trigger Metric	Push delivery success within 5 minutes falls below 99%, OR more than 5% of parent devices register invalid push tokens.
Contingency Management	Prioritise defect resolution for push adapters; send direct SMS notifications for critical circulars; release OEM battery-saver configuration guides.
Accountable Owner	Lead Developer (P1)

Risk 5: Messaging Rule Too Strict — Urgent Communications Delayed (Exposure Score: 12 — MEDIUM)

RMMM Dimension	Operational Protocol & Strategic Action Plan
Risk Mitigation	Implement streamlined emergency urgent route requiring Principal digital sign-off; default unapproved messages to 07:00 release; embed school office telephone numbers directly within the automated night auto-response.
Risk Monitoring	Weekly audit of urgent approval request counts, principal response latency, and parent complaint logs.
Trigger Metric	More than 2 parental complaints per week regarding delayed urgent matters, OR Principal approval latency exceeds 30 minutes.
Contingency Management	Principals designate an authorized vice-principal backup approver; management reviews sending window hours; urgent messages remain segregated in compliance audits.
Accountable Owner	Developer/QA (P3) with Pilot School Principals

7.4 Issue Log & Rationale for Risk-Issue Separation

Issue ID	Date Raised	Empirical Incident Description	Operational Impact	Corrective Remediation Action	Status	Root Risk
I-01	Day 25	186 of 2,300 parent contact numbers missing in pilot roster CSV	186 parents unreachable via push or SMS, capping initial adoption	Office staff collected numbers during orientation; admin portal added error filter	Open	R6 (Roster), R1
I-02	Day 23	Faculty training clash with scheduled terminal exams at School 2	Reduced faculty attendance and prep time at School 2 before go-live	Rescheduled session post-exams; distributed asynchronous video guides	Closed	R1 (Adoption)
I-03	Day 31	Push alerts suppressed by OEM battery saver on budget handsets	Delayed notice delivery for subset of parents; parent frustration	Published battery-saver setup guide; prioritized C-02 SMS Fallback	Open	R2 (Push failure)

Theoretical Justification — Why Risks and Issues Must Never Be Combined:

A **Risk** is an uncertain future event characterised by a probability ($0 < P < 1$) and potential impact, requiring proactive mitigation and threshold triggers. An **Issue** is a realized, certain event ($P = 1.0$) currently disrupting project execution, requiring immediate corrective action, an owner, and a closure target.

Combining both concepts into a single register destroys exposure scoring (a realized issue cannot be evaluated as $P \times I$) and causes urgent operational firefighting to crowd out long-term proactive risk mitigation. Maintaining separate logs illuminates how risks evolve into live issues (e.g., Risk R6 evolved into Issue I-01; Risk R2 evolved into Issue I-03).

SECTION 08

Project Closure & Retrospective Lessons

8.1 Pilot Results & Conditional Rollout Decision

Key Success Target	Target SLA	Actual Empirical Pilot Result	Variance & Contractual Status
Monthly Active Parents	90.0% (2,070 / 2,300)	1,978 / 2,300 parents = 86.0%	■ Shortfall of –4.0 pp (92 parents short); Not Met
Late Teacher Messages (>20:00)	0 messages	0 routine messages delivered after 20:00	■ 100% SLA Compliance; Met (3 approved urgent overrides)
Circular Delivery Timeliness (NFR-01)	≥99.0% within 5 min	99.3% delivered within 5 minutes	■ Exceeded SLA (+0.3 pp); Met
Defect Removal Efficiency (DRE)	≥90.0%	34 ÷ 36 = 94.4%	■ Exceeded Quality Target (+4.4 pp); Met
Scheduled Duration Finish	27 working days	29 working days (+2 working days)	Slip of +7.4% due to UAT defect remediation

EXECUTIVE GOVERNANCE DECISION: CONDITIONAL ROLLOUT TO REMAINING 10 SCHOOLS

The Project Steering Committee approves a **Conditional Rollout** to the remaining 10 institutions (11,700 parents). While technical SLA benchmarks (Zero late messages, 99.3% delivery timeliness, 94.4% DRE) were successfully achieved, parent adoption reached 86.0% against the 90.0% target. Consequently, expansion to the next school group is contingent upon: (1) fast-tracking C-02 SMS Fallback for non-smartphone households, and (2) achieving ≥80% adoption by Week 3 at each newly onboarded campus.

8.2 Lessons Learned (Engineering Retrospective)

#	Empirical Incident	Root Cause Analysis	Engineering / PM Lesson Learned	Corrective Policy for 10-School Rollout
1	Project completed on Day 29 (+2d slip beyond 27d baseline)	UAT defects discovered late; fixes executed inside critical testing window without contingency buffer	The critical path requires dedicated rework contingency buffers immediately preceding launch	Insert a mandatory 2-day defect remediation buffer post-UAT; initiate component UAT earlier
2	Adoption reached 86.0% vs 90.0% contractual target	Onboarding relied solely on faculty circulars; smartphone/data poverty in 8% of households	Parent engagement targets require dedicated socio-technical onboarding programs	Launch campus-wide onboarding weeks; fund and deploy C-02 SMS Fallback gateway
3	Messaging module required 14 days instead of 12 days	State machine rules, edge cases, and principal approval flows were more complex than estimated	Resource-levelled float is substantially tighter than CPM free float in multi-tasking teams	Add a 20% complexity contingency factor to rule-heavy modules; maintain visible float boards

#	Empirical Incident	Root Cause Analysis	Engineering / PM Lesson Learned	Corrective Policy for 10-School Rollout
4	DEF-35 (Midnight rollover dual delivery) escaped to pilot	TC-BVA-08/09 were executed on reset clocks; contiguous boundary rollover crossing 00:00 was never tested live	Time-dependent state engines require dynamic continuous clock boundary simulation	Introduce automated continuous clock-advance regression suites spanning midnight and month-ends
5	186 parent phone numbers missing at initial import (I-01)	Roster data sourced directly from unvalidated manual Excel sheets maintained by school offices	Data cleansing is a mandatory predecessor schedule task, not a benign project assumption	Enforce pre-pilot CSV data validation and dry-run syntax audits with formal admin sign-off
6	Push suppression on aggressive battery saver phones (I-03)	Testing executed primarily on modern developer handsets, masking legacy budget OEM aggressive power policies	Physical test matrices must match actual demographic hardware profiles, not modern devices	Integrate dedicated low-end Android 8 devices into test racks; publish OS-specific guidance

8.3 Formal Project Sign-Off Matrix

Stakeholder Role	Sign-Off Authority	Governance Decision / Scope Endorsement	Formal Status
Project Manager	Kishan Ojha	Closure Note, Traceability Matrix, and Engineering Metrics Certified	APPROVED & SIGNED
Group Management Sponsor	Representative	Conditional Rollout to Remaining 10 Institutions Endorsed	APPROVED (CONDITIONAL)
Quality Assurance Lead	Developer/QA (P3)	94.4% Defect Removal Efficiency and Test Logs Verified	CERTIFIED & ACCEPTED
School 1 Principal	Campus Representative	Pilot Operational Handover and Faculty Transition Confirmed	ACCEPTED
School 2 Principal	Campus Representative	Pilot Operational Handover and Faculty Transition Confirmed	ACCEPTED

SCHOOLDIARY · CASE STUDY #111 · FORMAL PROJECT REPORT

Sole Author: **Kishan Ojha** · B.Tech Computer Science & Engineering (2025–2029)

Academic Evaluation: Software Engineering & Project Management — Semester III

Comprehensive Engineering Deliverables: IEEE 830 SRS · UML Package · CPM Network Plan · Estimation Formulas · Scope

Boundary Matrix · IEEE 829 Test Suite · Risk Register & RMMM · Closure Note

All empirical pilot results, defect data, issue records, and interview statements are simulated academic artefacts constructed in strict adherence to the Case 111 problem specification.